



# GE and uncertainty

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# General equilibrium and uncertainty

Varios models.

1. Arrow-Debreu contingent claims. (Debreu 1959, chap. 7).
2. Arrow securities
3. Rational expectations equilibrium (REE). (Radner 1979).
4. General equilibrium, incomplete markets (GEI)
5. Dynamic variations
  - a. Dynamic completeness (DC)
  - b. Dynamic incompleteness (DI)

# Arrow-Debreu contingent-claims model

- Goods:  $\ell = 1, \dots, L$
- States:  $s \in \{1, \dots, S\}$
- There are two periods,  $t = 0$  and  $t = 1$ .
- $t = 0$ : agents are uncertain about  $s$ ,  $s$  is realized at  $t = 1$ .
- $t = 0$ : Markets for contingent claims open (complete markets) and agents trade them.
- A contingent claim  $x_{\ell s} \in \mathbb{R}_+$  is a claim to  $x_{\ell s}$  units of good  $\ell$  at time  $t = 1$  if state  $s$  is realized.
- At  $t = 1$ ,  $s$  is realized, contingent claims are exercised, and consumption takes place.

- Agent  $i$ :
  - $X_i \subseteq \mathbb{R}_+^{LS}$ ,
  - $\omega_i \in \mathbb{R}_+^{LS}$ ,  $\omega_{i\ell s}$  is the endowment of good  $\ell$  that agent  $i$  will have if state  $s$  occurs.
  - $\succeq_i$  is a binary relation on  $X_i$ .
- A state-contingent commodity bundle for  $i$  is

$$(x_{i11}, x_{i21}, \dots, x_{iL1}, x_{i12}, \dots, x_{iL2}, \dots, x_{i1S}, \dots, x_{iLS}) \in X_i,$$

- where  $X_i \subseteq \mathbb{R}_+^{LS}$ ; and
- $(x_{i1s}, x_{i2s}, \dots, x_{iLs})$  is  $i$ 's consumption bundle at  $t = 1$  if  $s$  occurs.

# Preferences

- $\succeq_i$  on  $X_i$  reflects the consumer's "tastes for goods and services (including, in particular, their spacial and temporal specifications), his personal appraisal of the likelihoods of the various events, and his attitude toward risk." (Debreu 1959, chap. 7, page 101).
- Convexity of preferences is naturally interpreted in the context of contingent claims as risk aversion.
- Common assumptions yield a utility representation for preferences,

$$x \succeq_i x' \iff u_i(x) \geq u_i(x').$$

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Additional assumptions yield specific representations:

- subjective probabilities and state-dependent expected utility

$$u_i(x) = \sum_{s=1}^S \pi_i(s) v_i(x_s, s) = \sum_{s=1}^S \pi_i(s) v_i(x_{1s}, \dots, x_{Ls}, s)$$

where  $v_i : \mathbb{R}^L \times \{1, \dots, S\} \rightarrow \mathbb{R}$

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# Results

- Formally identical to Arrow-Debreu economy: Goods and services are reinterpreted as contingent claims resulting in an Arrow-Debreu economy with  $LS$  goods.
- Then, existence of WE and welfare theorems hold under the same formal assumptions. Proofs remain the same.
- Implicit assumptions:
  - There are  $LS$  markets, one for each contingent claim.
  - One of the following alternative assumptions about the information structure:



# Alternative information assumptions

1. No private information: Every agent knows the true probability distribution over states, or
2. Common beliefs: Every agent agrees on a probability distribution over states, and maximizes expected utility with respect to the common beliefs—common beliefs may be wrong—or

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3. Private information: Agents have different beliefs about the distribution over states but do not update beliefs based on prices or actions of others.

Pareto optimality is in ex ante sense with respect to given beliefs: At  $t = 0$ , feasible allocation of contingent claims makes every agent better off given agents'  $t = 0$  beliefs.

# Arrow securities

- In the Arrow-Debreu contingent-claims model, before uncertainty is revealed, there is a market for every contingent claim to every commodity. This requires  $LS$  markets, an extraordinary number, and all trading takes place at  $t = 0$ .
- Innovation:
  - Introduce trading at  $t = 1$  (i.e. after  $s$  is revealed) for the  $L$  standard goods.
  - At  $t = 0$  allow trading of particular securities that suffice to transfer “income” between states.
- Essential model remains unchanged but the number of necessary markets is reduced.

# Features

- Full set of Arrow securities: one for each state  $s$ . An  $s$ -state Arrow security promises to pay one unit of account in state  $s$  and 0 for  $s' \neq s$ .
- At  $t = 0$ , securities are traded.
- At  $t = 1$ ,
- A state  $s$  is revealed. State- $s$  security pays 1 unit of account. Agents get a positive or negative dividend. The net dividend is zero.
- State- $s$  spot market opens, the  $L$  commodities and services are traded.
- Information assumptions: same as in Arrow-Debreu contingent claims model.

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- At  $t = 0$ , agent  $i$  may purchase  $z_i \in \mathbb{R}^S$  of Arrow securities at price  $q \in \mathbb{R}^S$  subject to budget constraint  $q \cdot z_i \leq 0$ .
  - If at  $t = 1$  the state  $s \in S$  is revealed,  $i$  faces the following budget constraint  $p_s \cdot x_{is} \leq p_s \cdot \omega_{is} + z_{is}$ .
  - An allocation of securities  $z = (z_1, \dots, z_I)$  is feasible if  $\sum_{i=1}^I z_i = 0$ .

# Results

**Theorem 1 (Arrow securities)** Let  $(x^*, p^*)$  be a WE of an Arrow-Debreu economy with contingent claims. If  $q = (1, \dots, 1)$ , and  $\forall i, s, z_{is} = p_s^* \cdot (x_{is}^* - \omega_{is})$ , then  $(x^*, z, p^*, q)$  is a WE with Arrow securities.

Conversely, let  $(x^*, z^*, p^*, q^*)$  be an equilibrium with Arrow securities. If  $\forall s, p_s = q_s^* p_s^*$ , then  $(x^*, p)$  is a WE with contingent claims.

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“*Theorem.*” A Walrasian equilibrium exists. If at  $t = 0$  agents correctly anticipate the spot-prices that prevail at  $t = 1$  in *every* state, the WE is Pareto optimal.

Note: It requires that agents correctly anticipate the spot prices at every state, not that they anticipate the state itself.

# Rational expectations equilibrium

- Asymmetric information.
- Agents infer from prices the information that other agents have.
- Rational expectations equilibria only exists generically.



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- $i = 1, \dots, I$  agents,  $\ell = 1, \dots, L$  goods
  - $i$ 's consumption set is  $X_i = \mathbb{R}_+^L$
  - $i$  has state dependent utility and endowments
  - $\Omega$  is the set of states of the world
  - $F_\Omega$  is a probability distribution on  $\Omega$
  - Information structure
    - Every agent  $i$  observes a signal  $s_i \in S^i$
    - Signal space  $S = \prod_{i=1}^I S^i$
    - $F$  is a probability distribution on  $S \times \Omega$  whose marginal on  $\Omega$  is  $F_\Omega$ .

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**Definition 1** A fully revealing rational expectations equilibrium is a price function  $\varphi : S \rightarrow \Delta$  and  $\forall p \in \varphi(S)$  a feasible allocation  $f_p : I \rightarrow \mathbb{R}_+^L$  such that

1.  $\varphi$  is one to one
2.  $f_p(i)$  maximizes

$$\left\{ \begin{array}{l} E[u_i(x, \omega) | s'_i, \varphi(s) = p] \text{ s.t.} \\ \bigcap_{s: \varphi(s)=p, s_i=s'_i} \{x : p \cdot x \leq p \cdot \omega(i, s)\} \end{array} \right.$$

# Assumptions and results

- Additional assumptions:  $S$  finite, or restrictions on the dimensionality of  $S$  relative to commodity space  $L$ . Smooth demands.
- Result: Generically (i.e. except for a measure zero set of economies parametrized by endowments), economies have a rational expectations equilibrium that fully reveals the joint signal profile  $s = (s_1, \dots, s_I)$ .

# Remarks

- Since  $\mathcal{S}$  is finite and  $\Delta$  has the cardinality of the continuum, it is easier to obtain a one-to-one function  $\varphi$ .

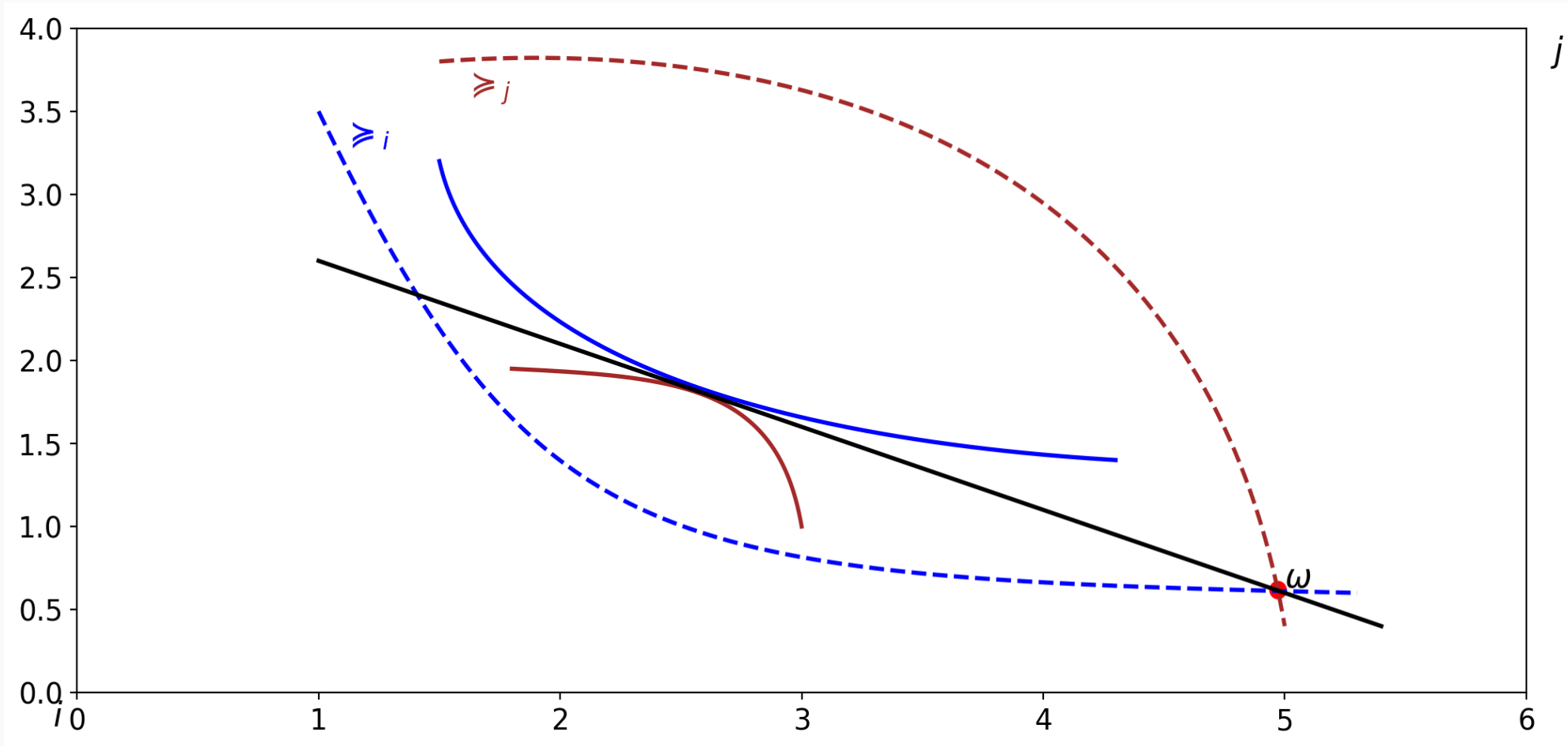
The correct model should have  $\dim \mathcal{S} > \dim \Delta$

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- How is the price  $\varphi(s)$  determined in the market?
    - If  $p \notin \varphi(S)$ , excess demand is not defined. Adjustment rule cannot depend on excess demand.
    - Suppose there is  $i$  with information that no other agent has, and this information does not affect  $i$ 's demand function. How do prices acquire this information? (Beja's critique)
    - If prices revealed all information, why would agents invest in acquiring information? (Grossman-Stiglitz)

# GE with incomplete markets

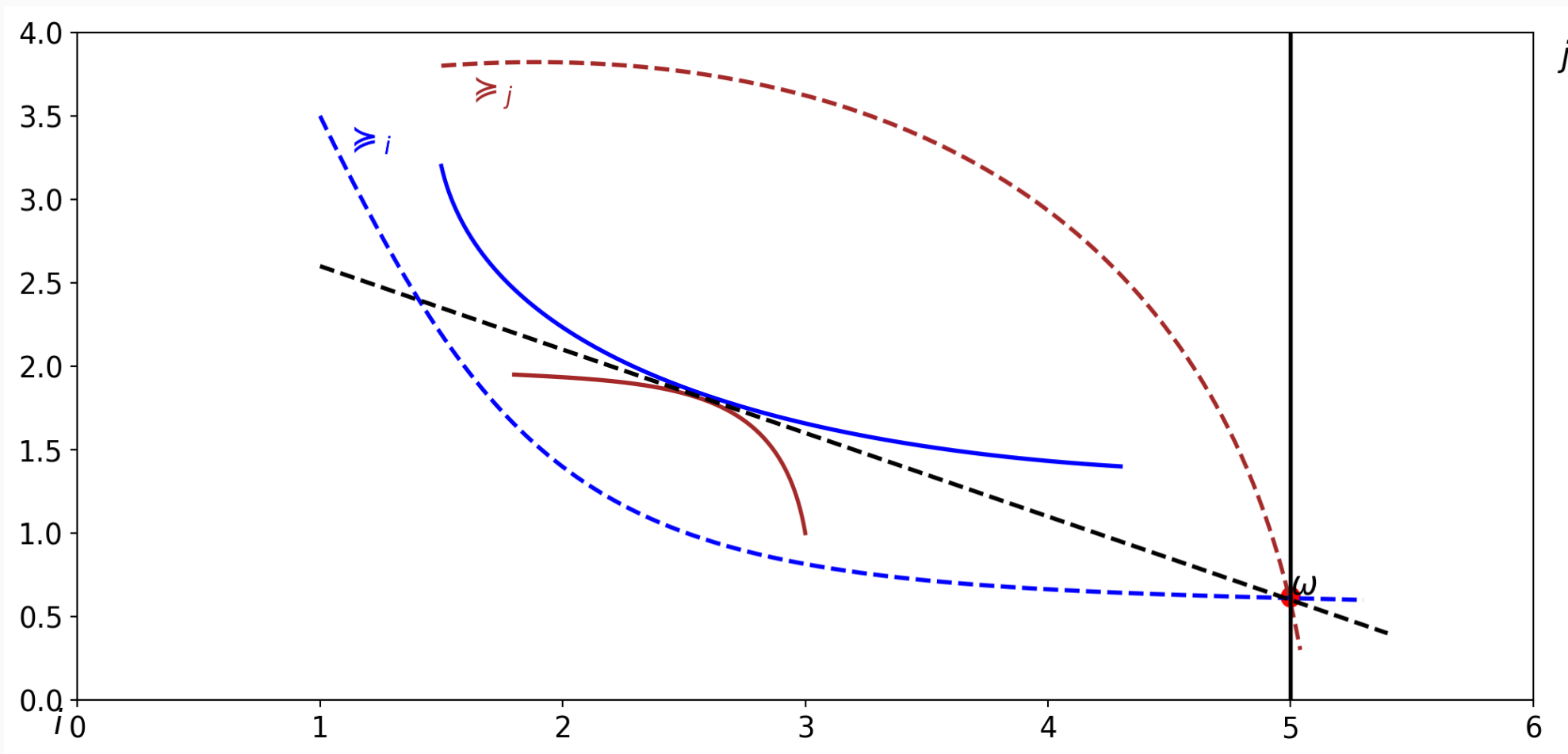
- Symmetric information.
- Two periods  $t = 0, 1$ ;  $L = 1$  good
- $s = 1, \dots, S$  states are revealed in period 1.
- $J$  securities,  $J < S$ , similar to Arrow securities but may pay in more than one state. Securities are traded at  $t = 0$ .
- Walrasian equilibrium exists (same arguments as in Arrow-Debreu economy).
- Both First and Second Welfare Theorems fail.

# Complete markets, two securities



- $L = 1, S = \{1, 2\}$ , 2 Arrow securities. For  $s \in S$ , security  $z_s$  pays 1 unit of account in state  $s$ .

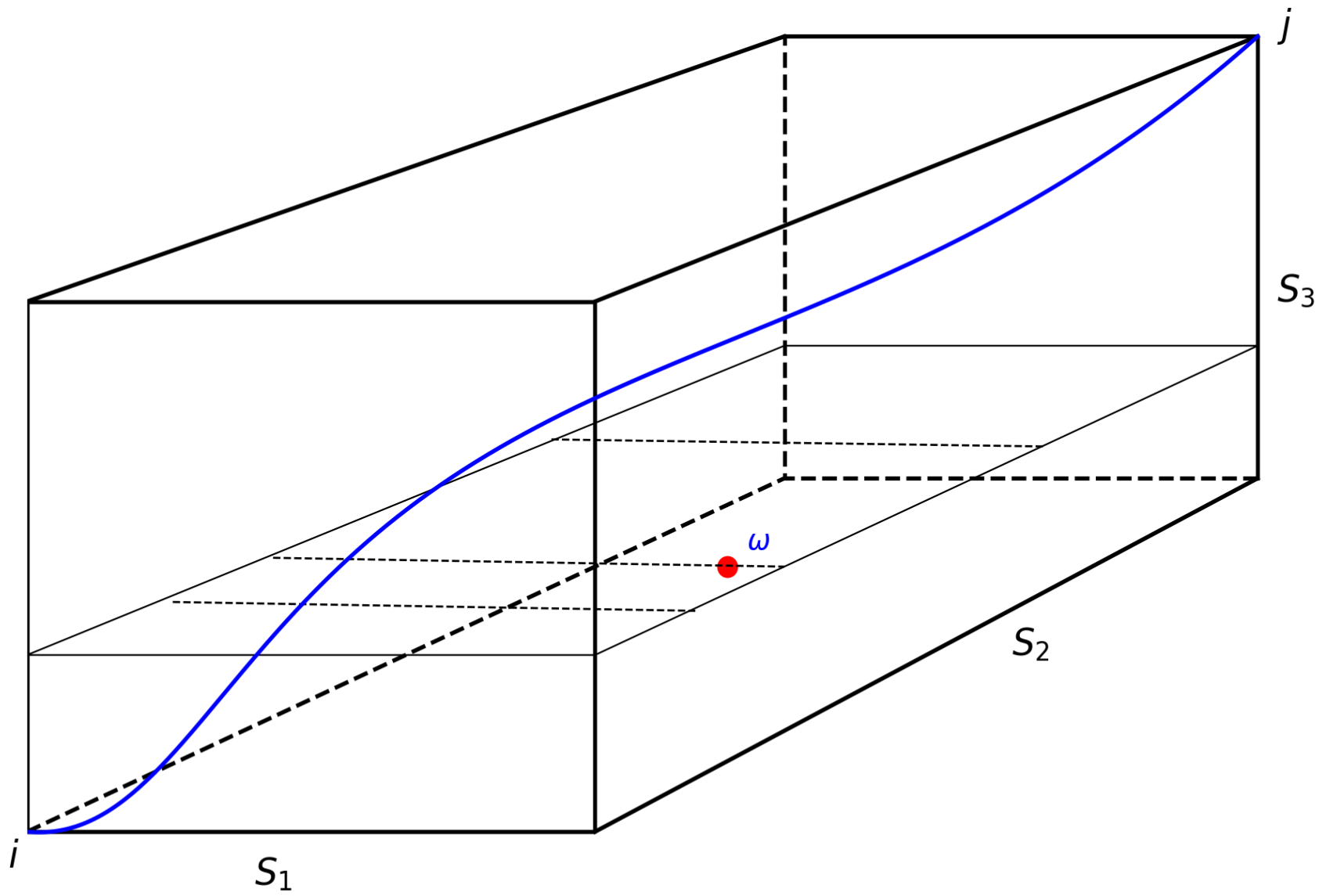
# Incomplete markets, one security



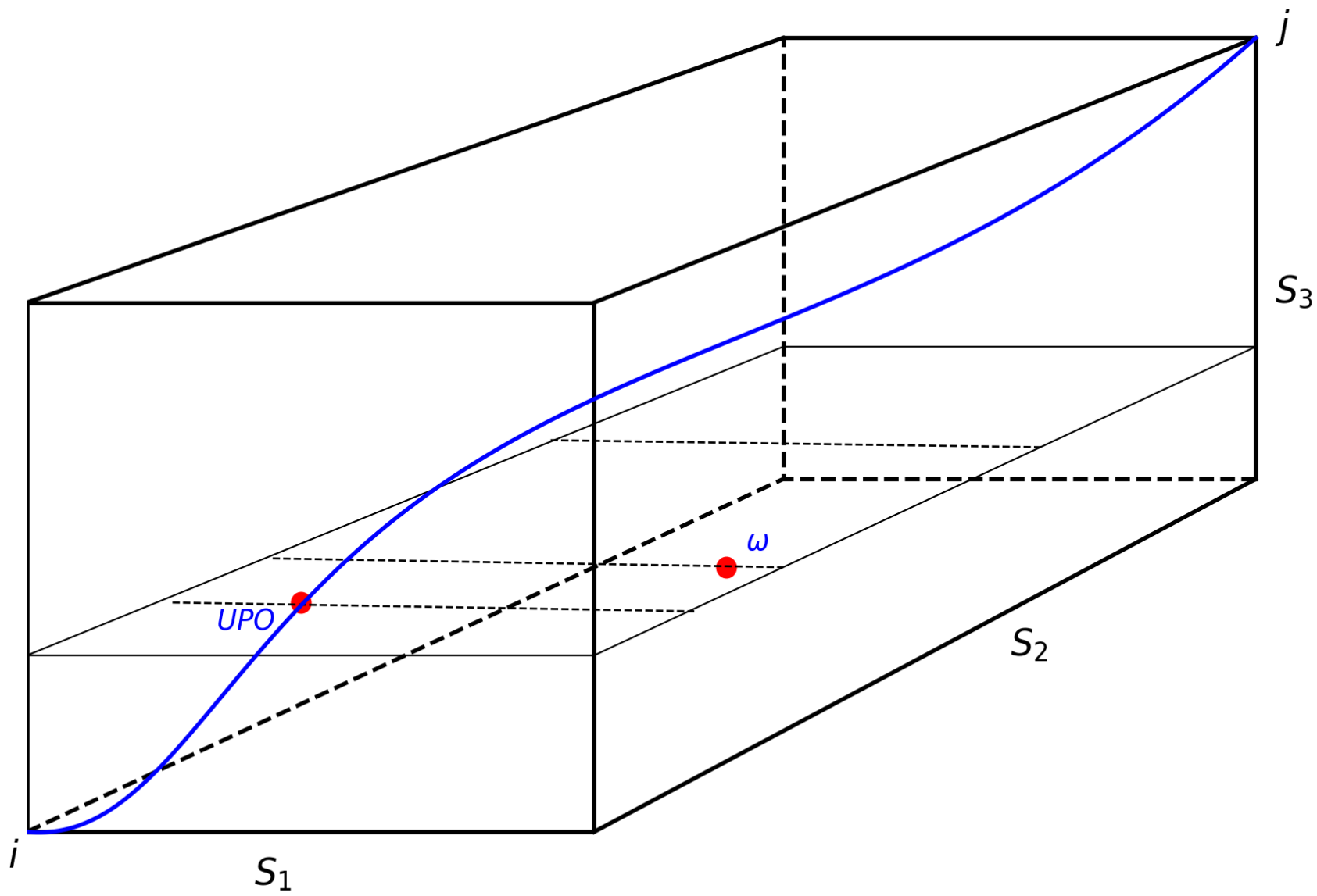
- $L = 1, S = \{1, 2\}$ , one Arrow security  $z_2$ .



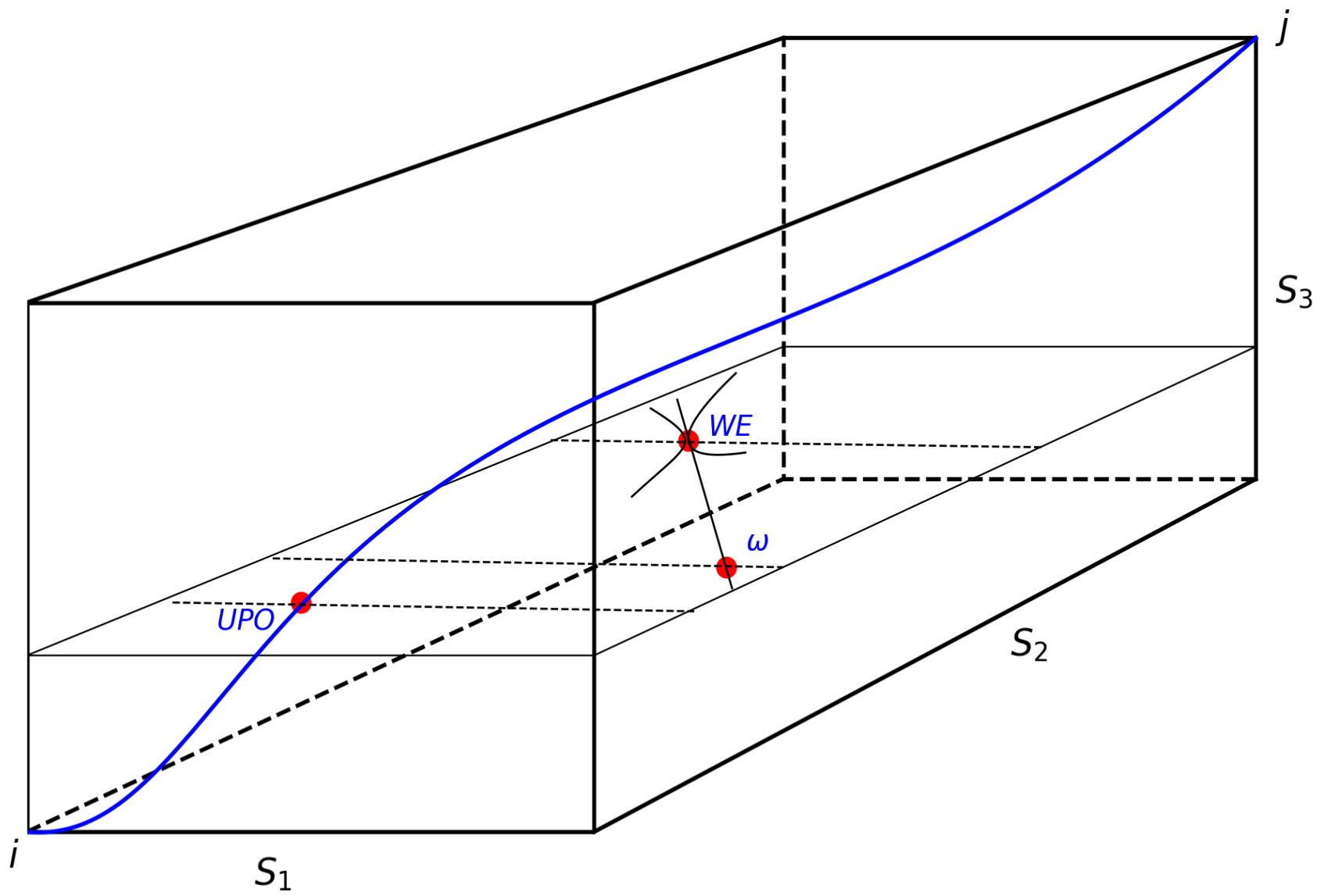
# Edgeworth cube 1



# Edgeworth cube 2



# Edgeworth cube



# Incomplete markets

- 2 periods,  $L > 1$  goods,
- Securities pay in goods.
- Or more than 2 periods,  $L = 1$  good.
- Conjecture. WE is constrained PO, there is no allocation achievable by the available markets that Pareto dominates the WE allocation.
- Note. Allocations achievable by given markets depend on prices.
- Example in (Hart 1975) and (Mas-Colell, Whinston, and Green 1995, 19.F.2).

# Hart example

- $I = 2, L = 2, S = 2$ . No securities.
- Agent  $i, E \succ_i E'$ ; Agent  $j E' \succ_j E$ .
- A consumption for agent  $i$  in state  $s$  of good  $\ell$  is  $x_{is\ell}$ . For given  $n \in \mathbb{N}$ , preferences are

$$u_i(x_{i11}, x_{i12}, x_{i21}, x_{i22}) = nv_i(x_{i11}, x_{i12}) + v_i(x_{i21}, x_{i22})$$
$$u_j(x_{j11}, x_{j12}, x_{j21}, x_{j22}) = v_j(x_{j11}, x_{j12}) + nv_j(x_{j21}, x_{j22})$$

- There are at least four equilibria,

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# References

Debreu, Gerard. 1959. *The Theory of Value*. New York: John Wiley.

Hart, Oliver D. 1975. "On the Optimality of Equilibrium When the Market Structure Is Incomplete." *Journal of Economic Theory* 11: 418–43. [https://doi.org/10.1016/0022-0531\(75\)90028-9](https://doi.org/10.1016/0022-0531(75)90028-9).

Mas-Colell, Andreu, Michael Whinston, and Jerry Green. 1995. *Microeconomic Theory*. New York, Oxford: Oxford University Press.

Radner, Roy. 1979. "Rational Expectations Equilibria." *Econometrica*.